



MEKELLE UNIVERSITY



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**Design and Evaluation of a Multi-Bank Transaction Verification
System with Intelligent Fraud Detection**

Thesis Project Proposal

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Submission Date Feb 15 2026 G.C

Contents

1. Abstract	4
1.1 Problem Statement.....	6
1.2 Objectives	6
1.2.1 General Objective	6
1.2.2 Specific Objectives	6
1.3. Research Questions.....	7
1.4. Scope and Limitations	7
1.4.1 Scope.....	7
1.4.2 Limitations.....	8
Chapter2: Literature Review	8
2.1 QR Code Authentication in Financial Systems	8
2.2 Fraud Detection Methodologies	9
2.3 Verification Interoperability and API Standards	9
2.4 Research Gap	9
Chapter 3. Methodology	10
3.1 Phase 1: Data Collection and Simulation.....	10
3.2 Phase 2: System Implementation.....	10
3.3 Phase 3: Experimentation and Testing.....	10
3.4. Proposed System Architecture	10
3.4.1 Input Layer.....	10
3.4.2 Preprocessing and QR Parsing Module	11
3.4.3 Bank Identification and Routing Module	11
3.4.4 Dual-Path Verification Engine	11
3.4.5 Output and Decision Layer.....	12
3.5 Intelligent Fraud Detection Model.....	12
3.6 Feature Engineering.....	13
3.7 Model Selection	13

3.8 Training and Evaluation Strategy.....	13
3.9 Evaluation Metrics.....	14
References.....	15

1. Abstract

The proliferation of digital banking in emerging economies has revolutionized financial inclusion, yet it has simultaneously introduced novel vectors for financial fraud. In Ethiopia, the rapid adoption of diverse digital payment platforms—such as those offered by Commercial Bank of Ethiopia (CBE), Dashen Bank, and Awash Bank—has outpaced the development of unified verification infrastructure. While some financial institutions provide verification endpoints, merchants frequently lack a centralized tool to access them, forcing reliance on static digital artifacts like screenshots or PDF receipts. This fragmentation creates a "verification gap" vulnerable to manipulation, including tampered QR payloads, reused transaction IDs, and counterfeit receipts.

This research proposes the design, implementation, and evaluation of a **Multi-Bank Transaction Verification System (MBTVS)** featuring a **hybrid validation architecture**. The system functions as a unified gateway that utilizes computer vision to parse Quick Response (QR) codes and extract structured transaction metadata. Based on the identified issuing institution, the system dynamically routes the verification request through one of two pathways: (1) a **Deterministic Verification Module**, which interacts with available banking APIs to validate transaction authenticity against the ledger in real-time; or (2) a **Probabilistic Intelligent Fraud Detection Engine** for banks lacking public APIs. The latter employs unsupervised anomaly detection algorithms and rule-based logic to identify reused IDs, temporal anomalies, and structural irregularities indicative of fraud. The system will be evaluated using a combination of live API integration testing and comprehensive synthetic datasets representing varying fraud scenarios. The expected contribution is a scalable, interoperable middleware that enhances trust in the digital economy by bridging the gap between direct API verification and offline receipt analysis.

CHAPTER 1: INTRODUCTION

The global financial landscape is undergoing a paradigm shift driven by the digitization of payment systems. In developing economies, this transition is characterized by the leapfrogging of traditional card-based infrastructures in favor of mobile-centric banking solutions. In Ethiopia, the modernization of banking infrastructure has led to an exponential increase in digital transaction volumes. Mobile banking applications have become the primary interface for retail transactions, peer-to-peer transfers, and merchant payments.

However, the architecture of trust in this ecosystem remains fragmented. While tier-one financial institutions have begun exposing Application Programming Interfaces (APIs) for transaction validation, these tools are often inaccessible to small merchants due to technical complexity and fragmentation. Unlike Point-of-Sale (POS) systems that receive immediate confirmation from a centralized gateway, many Small to Medium-sized Enterprises (SMEs) and informal merchants cannot integrate multiple disparate banking APIs. Consequently, they revert to "visual verification," where customers present a digital receipt—often a screenshot or a downloadable PDF—as proof of payment. This creates a disconnect between the transaction initiation (on the customer's device) and verification (by the merchant), resulting in a "verification gap."

Fraudsters exploit this gap through social engineering and technical manipulation. Common attack vectors include the alteration of payment amounts in image editors, the replay of valid QR codes from previous transactions, and the generation of entirely synthetic receipts that mimic legitimate banking formats. The absence of a unified validation mechanism exacerbates this issue; a merchant accepting payments from multiple banks faces the operational unfeasibility of managing distinct verification protocols for each provider.

This is motivated by the urgent need to bridge the gap between fragmented banking standards and the requirement for a unified, secure verification protocol. By aggregating direct API validation for supported banks and applying intelligent machine learning analysis for others, this study aims to secure the "last mile" of digital payments. The proposed **hybrid framework** ensures deterministic accuracy where APIs are available, while providing robust probabilistic fraud detection for receipts that must still be verified offline.

1.1 Problem Statement

Despite the ubiquity of digital payments, the ecosystem lacks a standardized, multi-bank mechanism for the offline or near-real-time verification of transaction receipts. The current state of practice relies heavily on manual human inspection of digital screens, which is inherently error-prone and vulnerable to sophisticated manipulation.

Specifically, the problem is threefold:

1. **Heterogeneity of Standards:** Ethiopian banks utilize disparate data structures, QR encoding formats, and verification URLs. There is no interoperable middleware that allows a third party to validate a receipt without manual intervention or specific knowledge of that bank's unique format.
2. **Vulnerability to Artifact Manipulation:** Static images (screenshots) lack cryptographic non-repudiation. Adversaries can easily modify transaction timestamps, amounts, and beneficiary details, or present a valid QR code from a past transaction (Replay Attack) to deceive merchants.
3. **Absence of Intelligent Anomaly Detection:** Current verification attempts are binary (valid/invalid) based on simple checks. There is a lack of systems capable of analyzing behavioral patterns. Therefore, there is a critical need for an automated system that acts as a unified verification layer, capable of parsing heterogeneous banking formats and employing intelligent algorithms to detect fraud in digital transaction receipts.

1.2 Objectives

1.2.1 General Objective

The primary objective of this research is to design, develop, and evaluate a **Hybrid Multi-Bank Transaction Verification System (H-MBTVS)**. This system will serve as a unified gateway that orchestrates transaction validation by integrating banking **APIs** for supported institutions (deterministic verification) and employing an **Intelligent Fraud Detection Engine** (probabilistic verification) to identify counterfeit receipts when API connectivity is unavailable or unsupported.

1.2.2 Specific Objectives

1. To develop a robust **QR Parsing and Data Normalization Module** capable of decoding, extracting, and standardizing transaction metadata from diverse banking receipt formats (e.g., CBE, Dashen, Awash) into a unified internal data structure.

2. To design a **Bank Identification and Routing Classifier** that utilizes pattern recognition to identify the issuing institution and dynamically route the verification request to either the API Gateway or the Fraud Detection Engine.
3. To implement a **Unified API Connector** that interfaces with available verification endpoints of supported banks to perform authentication of transaction references.
4. To construct an **Intelligent Fraud Detection Model** using unsupervised machine learning to calculate fraud risk scores for non-API transactions based on anomalies in timestamps, payload structure, and reuse patterns.

1.3. Research Questions

1. How can heterogeneous transaction data from diverse banking sources—ranging from static QR code payloads to dynamic API responses—be normalized into a unified data schema to support a single verification interface?
2. What is the optimal hybrid decision logic for routing verification requests between real-time banking APIs (where available) and the intelligent fraud detection engine (for unsupported or offline scenarios)?
3. To what extent can unsupervised machine learning algorithms (e.g., Isolation Forest, One-Class SVM) serve as an effective fallback mechanism for detecting fraud when direct API verification is unavailable or latent?

1.4. Scope and Limitations

1.4.1 Scope

- **Verification Modality:** The system is strictly focused on the verification of QR codes embedded in digital transaction receipts and the validation of their metadata via available external interfaces.
- **Institutional Coverage:** The study covers receipt formats and API structures resembling major Ethiopian financial institutions (e.g., CBE, Dashen, Awash) to demonstrate the system's multi-bank interoperability.
- **Hybrid Validation Logic:** The scope includes the development of a routing logic that switches between **Deterministic API Validation** (for supported banks) and **Probabilistic ML-based Detection** (for unsupported or offline scenarios).
- **Fraud Detection Boundaries:** Fraud detection is limited to content-based analysis (e.g., tampering, entropy, structural anomalies) and history-based analysis (e.g., ID

reuse). It does not extend to user behavioral biometrics, network-level forensic analysis, or identity verification (KYC).

1.4.2 Limitations

- **API Accessibility and Simulation:** While the system integrates with real banking APIs where permissioned access is available, it remains limited by the proprietary nature of financial interfaces. For banks that do not offer public developer sandboxes, the system will utilize **Mock APIs** and simulated REST endpoints that replicate the expected behavior and data structures of actual banking ledgers.
- **Environment Constraints:** The system will not have "Write" access to any banking database; it is a read-only verification tool. It operates as a middleware and does not facilitate the initiation or reversal of actual financial transactions.
- **Format Volatility:** The accuracy of the Parsing Module is dependent on the stability of bank receipt designs. Should an institution fundamentally alter its QR encoding logic or PDF layout, the system's parsing rules would require manual updates or retraining of the bank identification model.
- **Optical and Hardware Constraints:** The effectiveness of the initial data extraction is bounded by the quality of the input image. Severely blurred, low-resolution screenshots, or receipts with significant "noise" (e.g., heavy watermarking or glare) may fail the preprocessing stage, representing a limitation of the computer vision component.
- **Network Dependency:** The API-based verification path is subject to network latency and the uptime of third-party banking servers. In instances of downtime, the system must fall back to ML-based detection, which—while robust—is probabilistic and not 100% deterministic.

Chapter2: Literature Review

2.1 QR Code Authentication in Financial Systems

Quick Response (QR) codes have become the ubiquitous standard for proximity payments in emerging markets due to their high data capacity and error correction capabilities.

Tiwari et al. [1] highlight the critical distinction between static and dynamic QR codes, noting that static QRs—which are frequently used in generated transaction receipts—are vulnerable to (QR phishing) and replay attacks. While cryptographic signatures (e.g., PKI)

can theoretically secure these artifacts, most receipts generated in the local context lack digital signatures verifiable by third parties [2]. This creates a security gap where the verification relies solely on the visual content of the receipt, necessitating robust content-based verification approaches in the absence of real-time cryptographic validation.

2.2 Fraud Detection Methodologies

Financial fraud detection has historically relied on expert systems and deterministic rule-based logic. However, as noted by Phua et al. [3], these systems are rigid and often fail to detect novel or evolving attack patterns.

- **Supervised Learning:** Algorithms such as Logistic Regression and Random Forest have demonstrated high efficacy when labeled datasets of fraud exist [4]. However, obtaining labeled fraud data in the financial sector is notoriously difficult due to privacy concerns and the "cold start" problem for new attack vectors.
- **Unsupervised Learning:** Consequently, recent literature favors unsupervised approaches for anomaly detection. Algorithms such as the **Isolation Forest**, introduced by Liu et al. [5], and **One-Class Support Vector Machines (SVM)** are particularly effective in scenarios where fraud is a rare event (class imbalance). These models learn the statistical distribution of "normal" transactions and flag deviations, making them ideal for a hybrid system where the "normal" behavior of a valid receipt can be learned without explicit fraud labels.

2.3 Verification Interoperability and API Standards

The challenge of multi-bank interoperability is increasingly addressed through Open Banking protocols and API gateways [6]. In mature fintech ecosystems, verification is handled via direct calls to the issuing bank's ledger. However, in developing digital economies, the adoption of public verification APIs is fragmented. While some institutions have begun exposing RESTful endpoints for transaction validation, many rely on legacy infrastructure. In the absence of universal API coverage, researchers have explored Optical Character Recognition (OCR) and pattern matching as necessary "bridge technologies" [7]. Studies on digital document forensics demonstrate that inconsistencies in font rendering, compression artifacts, and metadata alignment can serve as proxy features for detecting tampered documents when direct API verification is unavailable [8].

2.4 Research Gap

Existing literature extensively covers credit card fraud detection using transactional logs and network security. However, there is a distinct paucity of research addressing the **forensic verification of digital transaction receipts** in a fragmented banking environment. Most studies focus exclusively on either *fully API-based* Open Banking

architectures or *fully offline* image forensics. Few studies propose a **Unified Hybrid Architecture** that orchestrates a dynamic handover between direct API verification (for supported banks) and intelligent machine learning analysis (for unsupported banks). This thesis addresses this specific gap by designing a system that secures the "merchant-view" of a transaction across multiple non-integrated banking providers, ensuring trust regardless of the underlying bank's technological maturity.

Chapter 3. Methodology

This research adopts the **Design Science Research (DSR)** methodology, involving the iterative development of an artifact (the software system) to solve a practical problem.

3.1 Phase 1: Data Collection and Simulation

- Development of a **Transaction Generator Script**. This Python script will reverse-engineer observed receipt formats from major Ethiopian banks (CBE, Dashen, Awash) to generate thousands of synthetic QR codes.

3.2 Phase 2: System Implementation

- **Language:** Python 3.9+.
- **Web Framework:** Flask or FastAPI.
- **ML Libraries:** Scikit-learn (for Isolation Forest/Random Forest), Pandas, NumPy.
- **Image Processing:** OpenCV and PyZbar.

3.3 Phase 3: Experimentation and Testing

- **Unit Testing:** Validating individual parsers for each bank.
- **Integration Testing:** End-to-end flow from image upload to risk scoring.

3.4. Proposed System Architecture

The system follows a modular, microservices-based pipeline architecture designed for scalability, interoperability, and high availability. The architecture is divided into five core layers.

3.4.1 Input Layer

The system accepts multi-modal input to accommodate various merchant capabilities:

- **Static Image Upload:** Processing of JPEG/PNG files containing digital transaction receipts.
- **Real-time Stream:** Frame capturing from a mobile camera feed for instant scanning.
- **Raw Payload Input:** Direct entry of transaction strings (if the QR is unreadable).

3.4.2 Preprocessing and QR Parsing Module

This module is responsible for normalizing raw visual data into a structured format.

- **Image Preprocessing:** Application of grayscale conversion, adaptive Gaussian thresholding, and morphological operations to correct skew and enhance QR code contrast.
- **Decoding:** Utilization of computer vision libraries to locate and decode the QR payload.
- **Data Normalization:** The raw payload—whether a URL, JSON, or unstructured string—is parsed and mapped to a **Unified Transaction Object (UTO)**. This ensures that downstream modules receive a standard data structure regardless of the originating bank (e.g., CBE, Dashen, Awash).

3.4.3 Bank Identification and Routing Module

This module acts as the system's traffic controller.

- **Classification Logic:** A hybrid classifier determines the issuing institution using URL domain patterns (e.g., *.cbe.com.et), payload schemas, and metadata keywords.
- **Routing Controller:** Once the bank is identified, the system queries an internal **API Capability Map**.
 - *Route A:* If the bank supports external verification, the request is routed to the **API Verification Gateway**.
 - *Route B:* If the bank does not have a public API, the request is routed to the **Intelligent Fraud Detection Engine**.

3.4.4 Dual-Path Verification Engine

3.4.4.1 Path A: API Verification Gateway (Deterministic)

For supported banks, this module executes a "hard" verification against the financial institution's ledger.

- **Connector Service:** Manages secure authentication and connection pooling.

- **Ledger Query:** Sends the extracted Transaction ID and Reference Number to the bank's endpoint.
- **Response Parsing:** Interprets the bank's response codes (e.g., 200 OK, 404 Not Found) to confirm existence and amount matching.

3.4.4.2 Path B: Intelligent Fraud Detection Engine (Probabilistic)

For unsupported banks, this module employs a "soft" verification using heuristic and machine learning analysis.

- **Syntax Validation:** Checks if the extracted data adheres to the specific regex patterns of the identified bank.
- **Anomaly Detection (ML):** The extracted features (timestamps, entropy, amount distribution) are fed into the **Isolation Forest** model to detect outliers indicative of image tampering or synthetic generation.

3.4.5 Output and Decision Layer

The system aggregates the results from either path into a standardized JSON response:

- **Verification Source:** Indicates whether the result came from Bank_API or Internal_ML_Engine.
- **Status:** Verified (Confirmed by API), Low_Risk (Passed ML checks), Suspicious (ML Anomaly), or Invalid (Failed syntax/API).
- **Risk Score:** A normalized value (0–100) quantifying the confidence of the verification.
- **Extracted Metadata:** The parsed amount, date, and payer name for visual confirmation by the merchant.

3.5 Intelligent Fraud Detection Model

The Intelligent Fraud Detection Model represents the core probabilistic component of the hybrid architecture. While the API Gateway provides deterministic validation for supported institutions, this module addresses the "Verification Gap" for banks lacking public endpoints or during instances of network latency. It employs a **Hybrid Logic** approach, fusing strict rule-based heuristics (for known patterns) with unsupervised machine learning (for unknown anomalies).

3.6 Feature Engineering

To enable the machine learning classifiers to distinguish between legitimate and fraudulent receipt artifacts, raw transaction metadata is transformed into a high-dimensional feature vector. Key features include:

- **ID Entropy and Complexity:** Calculation of the Shannon entropy of the Transaction ID strings. Fraudsters often generate sequential or low-entropy IDs that deviate from the pseudo-random generation algorithms used by banking systems (e.g., CBE or Dashen Bank core systems).
- **Temporal Consistency (Time-Delta):** The calculated difference between the timestamp embedded in the QR payload and the system's current scan time. Large deltas without API validation suggest a "Stale Receipt" or potential replay attack.
- **Payload Structural Integrity:** A measure of deviation from the expected standard. This includes checking payload length, delimiter positions, and parameter ordering specific to the identified bank's format.

3.7 Model Selection

The system utilizes an ensemble approach to handle the class imbalance inherent in fraud detection:

1. **Isolation Forest (Primary Anomaly Detector):** Selected for its efficacy in high-dimensional datasets where fraudulent instances are rare and unlabeled. Unlike distance-based methods, Isolation Forest explicitly isolates anomalies by randomly selecting a feature and a split value. Anomalies (fraudulent receipts) are susceptible to isolation closer to the root of the decision tree (shorter path length) compared to normal banking patterns.
2. **Logistic Regression (Calibration Layer):** Used as a secondary meta-learner to map the raw anomaly scores from the Isolation Forest and the binary outputs of the rule-based checks into a calibrated **Risk Probability Score (0.0–1.0)**. This score dictates the final decision (e.g., "Low Risk" vs. "Suspicious") presented to the merchant.

3.8 Training and Evaluation Strategy

- **Synthetic Data Injection:** Due to the sensitivity and scarcity of real-world fraudulent banking data, the research will employ a **Generative Adversarial approach**. A data simulation engine will generate a "Clean" dataset mimicking valid traffic from Ethiopian banks (CBE, Awash). A "Contaminated" dataset will then be created by injecting specific attack vectors:
 - *Replay Attacks:* Duplicates of valid IDs.

- *Fuzzing Attacks*: Randomly altered characters or amounts in the QR payload.
 - *Format Anomalies*: IDs that deviate from the bank's specific Regex patterns.
- **Training Protocol**: The Isolation Forest will be trained strictly on the "Clean" dataset to learn the manifold of normal transactions.
- **Testing Protocol**: The model will be evaluated against the "Contaminated" dataset to measure its ability to flag the injected anomalies as outliers.

3.9 Evaluation Metrics

Standard accuracy is an insufficient metric due to the "Accuracy Paradox" in fraud detection (where a model predicting "Valid" 100% of the time achieves high accuracy but zero utility). The evaluation will focus on:

- **Recall (Sensitivity)**: The system's ability to detect simulated fraud attempts (minimizing False Negatives, which represent financial loss).
- **Precision**: The proportion of flagged transactions that were truly fraudulent (minimizing False Positives, which cause customer friction).
- **F1-Score**: The harmonic mean of Precision and Recall, providing a balanced view of the model's performance.

References

- [1] S. Tiwari, "An introduction to QR code technology," *International Journal of Science and Research (IJSR)*, vol. 5, no. 7, pp. 2028–2030, 2016.
- [2] A. K. Jain and D. Shanbhag, "A review on QR code security: Challenges and solutions," in *2020 5th International Conference on Communication and Electronics Systems (ICCES)*, Coimbatore, India, 2020, pp. 1198–1202.
- [3] C. Phua, V. Lee, K. Smith, and R. Gayler, "A comprehensive survey of data mining-based fraud detection research," *arXiv preprint arXiv:1009.6119*, 2010.
- [4] Y. Sahin and E. Duman, "Detecting credit card fraud by decision trees and support vector machines," in *Proceedings of the International MultiConference of Engineers and Computer Scientists*, vol. 1, Hong Kong, 2011, pp. 442–447.
- [5] F. T. Liu, K. M. Ting, and Z. H. Zhou, "Isolation forest," in *2008 Eighth IEEE International Conference on Data Mining*, Pisa, Italy, 2008, pp. 413–422.
- [6] M. Zachariadis and P. Ozcan, "The API economy and digital transformation in financial services: The case of open banking," *SWIFT Institute Working Paper No. 2016-001*, 2017.
- [7] V. Kher, S. Brahmaxatriya, and P. Garg, "Forensic analysis of QR code based attacks," in *2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS)*, Coimbatore, India, 2021, pp. 1421–1426.
- [8] T. S. Worku, "Electronic-banking in Ethiopia: Practices, opportunities and challenges," *Journal of Internet Banking and Commerce*, vol. 15, no. 2, pp. 1–8, 2015.